

CONSTANTA, Romania

WMO: 154800

Lat: 44.2139N

Lon: 28.6456E

Elev: 14

StdP: 101.16

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.2	-6.8	-11.9	1.3	-8.4	-9.6	1.7	-5.5	14.3	-3.6	12.4	-1.5	4.7	0	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	30.4	23.3	29.2	22.8	28.1	22.3	25.5	28.3	24.3	27.5	23.3	26.8	2.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	19.7	27.3	23.2	18.0	26.5	22.2	16.9	25.7	78.3	28.4	73.3	27.6	69.6	26.9	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.4	6.7	-12.0	32.9	2.4	1.9	-13.7	34.3	-15.1	35.4	-16.5	36.4	-18.2	37.7		
(5)			-12.3	26.2	2.4	1.6	-14.1	27.3	-15.5	28.2	-16.8	29.1	-18.6	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	1.6	3.2	6.8	11.2	16.8	21.4	23.8	23.8	19.5	13.8	8.8	3.4		
	DBStd	8.65	4.83	4.78	3.84	3.24	2.98	2.63	2.16	2.27	2.94	3.65	4.43	4.61		
	HDD10.0	883	262	193	114	22	1	0	0	0	0	11	73	208		
	HDD18.3	2502	519	423	358	214	66	6	0	1	22	144	287	462		
	CDD10.0	1939	1	3	14	59	211	341	428	428	284	129	37	4		
	CDD18.3	517	0	0	0	1	18	97	170	170	56	5	1	0		
	CDH23.3	3206	0	0	1	7	49	494	1234	1242	171	8	1	0		
	CDH26.7	630	0	0	0	1	4	77	269	263	15	1	0	0		
(14) Wind	WSAvg	2.7	3.1	3.0	3.0	2.6	2.4	2.3	2.2	2.2	2.5	2.7	2.8	3.0		
(15) Precipitation	PrecAvg	444	35	29	32	31	42	42	36	32	37	37	42	40		
	PrecMax	708	155	98	96	109	135	145	125	138	150	108	116	188		
	PrecMin	225	4	2	0	2	3	2	0	0	0	3	0	2		
	PrecStd	107	28	22	23	21	30	30	29	32	34	27	32	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	17.5	21.3	23.6	26.9	30.6	32.5	32.1	28.4	24.2	21.2	15.9		
		MCWB	9.6	12.0	12.7	15.3	18.2	22.0	22.1	24.6	19.9	18.3	16.0	12.1		
	2%	DB	11.1	13.5	17.2	20.3	24.6	28.7	30.4	30.4	26.4	22.1	18.0	13.6		
		MCWB	8.3	9.7	11.1	14.1	18.1	21.8	23.3	23.8	20.4	17.9	14.8	11.2		
	5%	DB	9.3	11.1	14.4	18.0	23.0	27.3	29.2	29.2	25.1	20.4	16.2	11.5		
		MCWB	7.4	8.1	9.7	13.1	17.9	21.2	22.9	23.1	19.8	17.2	14.1	9.8		
	10%	DB	7.6	9.0	12.1	16.0	21.7	26.1	28.1	28.1	24.1	19.0	14.7	9.8		
		MCWB	6.3	6.9	8.6	12.3	17.2	20.7	22.5	22.6	19.6	16.0	12.7	8.4		
	0.4%	WB	10.3	12.3	13.7	16.3	20.7	24.7	27.3	27.0	23.1	20.3	16.7	12.8		
		MCDB	12.5	16.4	19.8	21.2	23.9	27.8	29.4	29.6	25.2	22.1	19.6	14.9		
	2%	WB	8.7	10.0	11.6	14.9	19.5	23.2	25.7	25.6	22.1	19.0	15.4	11.5		
		MCDB	10.6	12.9	16.2	19.1	22.8	26.9	28.2	28.6	24.4	20.9	17.5	13.1		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	7.5	8.5	10.1	13.7	18.6	22.4	24.4	24.4	21.2	17.7	14.3	10.0		
		MCDB	9.0	10.7	13.7	16.9	21.8	25.9	27.5	27.8	23.7	19.9	15.9	11.3		
	10%	WB	6.5	7.1	9.0	12.7	17.8	21.5	23.4	23.5	20.4	16.5	13.1	8.5		
		MCDB	7.6	8.8	11.6	15.4	21.0	25.0	26.9	26.9	23.2	18.6	14.3	9.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.8	6.3	6.2	6.7	7.1	7.3	7.2	6.9	6.2	5.7	5.6		
		MCDBR	7.7	9.4	10.6	9.8	9.0	8.5	8.5	8.1	7.8	8.0	7.6	7.7		
		MCWBR	5.5	6.0	5.9	5.1	4.8	4.8	4.9	5.1	4.5	4.9	4.8	5.6		
		MCDBR	7.1	8.7	9.4	8.2	7.8	7.3	7.1	7.3	6.8	6.9	7.0	7.1		
	5% WB	MCWBR	5.4	6.0	5.7	4.7	4.7	4.7	5.0	5.2	4.7	4.9	4.7	5.6		
(40) Clear-Sky Solar Irradiance	taub		0.353	0.382	0.412	0.447	0.424	0.425	0.437	0.447	0.424	0.395	0.372	0.353		
	taud		2.316	2.244	2.191	2.116	2.271	2.302	2.278	2.242	2.277	2.324	2.328	2.329		
	Ebn at Noon		746	789	815	816	850	848	832	808	795	766	717	704		
	Edh at Noon		84	107	128	149	131	128	130	130	116	97	82	76		
(44) All-Sky Solar Radiation	RadAvg		1.29	2.07	3.34	4.83	6.12	6.66	6.64	5.93	4.28	2.67	1.51	1.14		
	RadStd		0.19	0.27	0.34	0.47	0.37	0.38	0.34	0.31	0.33	0.25	0.19	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+109	+67
(47) Regional (26 neighbors)		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	-123	+97	+61

Nomenclature: See separate page